

KENYA JUNIOR SCHOOL EDUCATION ASSESSMENT



GRADE 9 2025

INTEGRATED SCIENCE PAPER 2 (PRACTICAL)

Time: 2 hours

Name:

School:

Assessment Number:

Signature: Date:

INSTRUCTIONS TO LEARNERS

1. You are supposed to work alone or in groups of 4 to 5 learners.
2. In case of any clarification, seek help from your subject teacher.

FOR OFFICIAL USE ONLY

Task	Maximum Score	Learner's Score
Task 1	15	
Task 2	15	

TASK 1 (15 MARKS)

1.

A learner was required to use solutions A, B, C and D as test solutions. The learner also used soap solution (solution X) to determine whether the test solutions were hard or soft water.

- (a) The learner added about 2 cm³ of soap solution (solution X) to about 5 cm³ of distilled water (soft water) and shook the test tube gently.
- (b) The learner added about 2 cm³ of soap solution (solution X) to about 5 cm³ of tap water (hard water) and shook gently.
- (c) The learner then added about 2 cm³ of soap solution to 5 cm³ of each of the test solutions A to D, one at a time, and recorded the lather formation or scum in the table below.

Complete the table by making appropriate conclusions: state whether the test solution is hard or soft water. (6 marks)

Substance	Observation	Conclusion
Soap solution + Distilled water	Formed lather immediately	
Soap solution + Tap water	Scum formed, little/no lather	
Soap solution + Test solution A	Scum formed	
Soap solution + Test solution B	Lather formed quickly	
Soap solution + Test solution C	Lather formed	
Soap solution + Test solution D	Little lather, some scum	

- (d) Name two basic science skills necessary to carry out this practical. (2 marks)

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- (e) Name three laboratory apparatus necessary for this practical. (3 marks)

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- (f) Name one water source that could be used in place of:

- (i) Distilled water (1 mark)
- (ii) Tap water (1 mark)

- (g) State two safety precautions that should be taken during this practical. (2 marks)

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TASK 2 (15 MARKS)

2.

Your class visited a nearby field to study the interdependence of living and non-living things. You observed grass, insects, birds, water, stones, and saw a farmer spraying chemicals on crops. Use this observation to answer the following questions.

(a) Classify the things you observed during the field study as **biotic** or **abiotic**. **(5 marks)**

Item	Biotic/abiotic

(c) State two ways in which water supports living things in the environment. **(2 marks)**

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(d) Outline two effects of the farmer’s spraying activity on the environment. **(2 marks)**

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(e) State two ways in which animals depend on plants. **(2 marks)**

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(f) List two safety rules you should follow during a field practical. **(2 marks)**

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CONFIDENTIAL — MARKING SCHEME

Keep this document confidential.

1 — TASK 1 (Total 15 marks)

A. Model Answers (directly fillable into the candidate's table)

Substance	Observation	Conclusion (marking)
Soap solution + Distilled water	Formed lather immediately	Soft water. (1 mark)
Soap solution + Tap water	Scum formed, little/no lather	Hard water. (1 mark)
Soap solution + Test solution A	Scum formed	Hard water. (1 mark)
Soap solution + Test solution B	Lather formed quickly	Soft water. (1 mark)
Soap solution + Test solution C	Lather formed	Soft water. (1 mark)
Soap solution + Test solution D	Little lather, some scum	Hard water. (1 mark)

(Total for table = 6 marks; 1 mark per correct conclusion in the “Conclusion” column. Observation column is taken from the question and does not need to be re-marked.)

B. Marking Scheme — Question 1 (breakdown)

(c) Table conclusions — **6 marks** (1 mark each for correctly stating soft/hard for Distilled, Tap, A, B, C, D).

(d) Two basic science skills — **2 marks**

1 mark each. Accept any two of: *Observation*; *Recording/Data recording*; *Measurement (accurate measurement of volumes)*; *Use of apparatus (handling pipette/test tube)*; *Inference/conclusion drawing*; *Safe handling*.

Model answer: Observation; Recording/Recording results. (Give full 2/2 if either pair is present.)

(e) Three laboratory apparatus — **3 marks**

1 mark each. Accept any correct apparatus used in this practical. **Expected:** Test tubes (or test-tube + rack), Pipette/dropper or measuring cylinder, Beaker (or measuring cylinder), Test-tube rack, Stirring rod, Labels/marker. (Award 1 mark per correct apparatus listed.)

(f)(i) Alternative for distilled water — **1 mark**

Accept: Deionised water, boiled & cooled water filtered (not tap), laboratory-grade purified water. (Model: Deionised water.)

(f)(ii) Alternative for tap water — **1 mark**

Accept: Well water, river water, borehole water (i.e., other natural water likely to be hard). (Model: Borehole water.)

(g) Two safety precautions — **2 marks**

1 mark each. Accept: Wear goggles; Wear gloves; Do not taste/ingest liquids; Handle glassware carefully to avoid breakage; Wash hands after the test; Clean spills immediately.

Total Task 1 = 15 marks.

C. Apparatus & Materials (Task 1 — recommended list and quantities)

- i. Soap solution X — prepared (see Solutions section). **Prepare at least 100 mL.**
- ii. Distilled (or deionised) water — **≥ 50 mL.**
- iii. Tap water (or alternate) — **≥ 50 mL.**
- iv. Test solutions A, B, C, D — **5 mL each** (as supplied by instructor).
- v. Test tubes — **minimum 8** (one for distilled, one for tap, four for A–D, one blank/control, spare).
- vi. Test-tube rack — 1.
- vii. Droppers or pipettes (1 mL or graduated) — enough to deliver 2 cm³ (2 mL) accurately; **at least 1 per student/group.**
- viii. Measuring cylinder (10 or 25 mL) — 1.
- ix. Beaker (100 mL) — 1 (for preparing/holding soap solution).
- x. Labels/marker and permanent marker tape.
- xi. Safety goggles and disposable gloves — 1 pair per student.
- xii. Waste container / cloth for spills.
- xiii. Stirring rod.

D. Solutions & How to Prepare Them (explicit)

Soap solution (Solution X) — standard classroom preparation (1% w/v liquid soap equivalent):

- Grated pure soap or commercial liquid soap may be used. Two options:

Option A (from bar soap)

1. Grate 1.0 g of soap into a beaker.
2. Add ~100 mL warm distilled water.
3. Stir until dissolved; cool and decant/filter if necessary.
→ **Result:** ~1% (w/v) soap solution. Use 2 mL per test.

Option B (from liquid hand soap)

4. Mix **5 mL liquid soap** with **95 mL distilled water** → total 100 mL (≈5% v/v). This is stronger; teacher may advise dilution to ~1–2% depending on soap strength.
5. Mix gently; label clearly.

Notes on concentration and usage: 2 cm³ (2 mL) of the soap solution added to 5 cm³ (5 mL) water/sample is the procedure used in the practical — maintain these volumes for consistency.

E. Procedure (step-by-step for the examiner's instructions and for candidates)

(Set up and checks before candidates begin)

1. Label test tubes: D (Distilled), T (Tap), A, B, C, D-test (or D-solution), and control/spare. Place in rack.
2. Provide 5 mL of distilled water in tube D, 5 mL tap water in tube T, and 5 mL of each test solution A–D in their tubes. Provide soap solution X in a small beaker/dropper.

Candidate procedure (steps for learners):

1. Put on goggles and gloves.
2. Using a pipette or dropper, add **2 cm³ (2 mL)** of soap solution X to **5 cm³ (5 mL)** of distilled water in tube labeled D. Replace stopper and gently invert/shake tube ~10 seconds. Observe and record whether a stable lather forms or scum appears.
3. Repeat step 2 with tap water in tube T. Observe and record.
4. Repeat step 2 for test solutions A, B, C, D — one test tube at a time. Record observations (lather quickly / little lather / scum).
5. Clean up: pour liquids into designated waste container (not down sink if contaminated), rinse glassware, remove gloves and wash hands. Return apparatus.

Observation timing guidance: wait ~10–30 seconds after shaking to assess lather stability (if lather persists it indicates soft water; scum or immediate collapse indicates hardness).

F. Safety Precautions (model answers to award marks)

- Wear eye protection (goggles) and gloves. (1 mark)
- Do not taste or ingest test solutions. (1 mark)
- Handle broken glass carefully; report to teacher. (award if mentioned)
- Dispose of test liquids as instructed by teacher.

G. Examiner's Notes — Common Errors & Deductions

- Wrong conclusion for a solution (soft vs hard) — **0/1 for that item**.
- If candidate writes only observation but no conclusion — award observation credit? (Observation is given in question; conclusion required — 0 for conclusion).
- Incorrect alternative water sources — zero for that item.
- Missing two safety rules — partial credit allowed (1/2 if one correct).
- If candidate uses wrong volumes but states correct trend, award partial credit for trend (examiner discretion).

2 — TASK 2 (Field Study) (Total 15 marks)

A. Model Answers

(a) Classification — 5 marks (1 mark each)

Use the items the candidate observed: grass, insects, birds, water, stones, farmer spraying chemicals. (If the question provided exactly six items, mark each classification 1 mark.)

Expected classification:

- i. Grass — **Biotic** (1 mark)
- ii. Insects — **Biotic** (1 mark)
- iii. Birds — **Biotic** (1 mark)
- iv. Water — **Abiotic** (1 mark)
- v. Stones — **Abiotic** (1 mark)

(If “farmer spraying chemicals” is listed as an item instead of farmer/chemicals separately, accept “Farmer — biotic” and “Chemicals — abiotic” with appropriate split by examiner.)

(b) Food chain — 2 marks

- **Model:** Grass → Insect → Bird. (Award 2/2 for a correct simple food chain showing correct direction.)
- Acceptable variants: Grass → Grasshopper → Bird; Grass → Insect → Small Bird.

(c) Two ways water supports living things — 2 marks

(1 mark each) Accept any two of: *Transport of nutrients and substances in plants and animals; habitat for aquatic organisms; used in photosynthesis; regulates temperature; drinking source.*

Model answers: Used in photosynthesis by plants; Drinking resource for animals.

(d) Two effects of farmer's spraying activity — 2 marks

(1 mark each) Accept: *Kills pests (positive) but may kill beneficial insects (negative); chemical residues can contaminate water sources; bioaccumulation in food chain; soil pollution; human health risk.*

Model: Kills useful insects (e.g., pollinators); contaminates nearby water, harming aquatic organisms.

(e) Two ways animals depend on plants — 2 marks

(1 mark each) Accept: *Food (herbivores), shelter (nesting), oxygen production, seeds/fruit for dispersal, materials (e.g., nesting material).*

Model: Animals eat plants; Animals use plants for shelter/nesting.

(f) Two safety rules during field practical — 2 marks

(1 mark each) Accept: *Wear suitable footwear/clothing; do not touch unknown plants or chemicals; stay with group; avoid drinking local water; report hazards.*

Model: Wear closed shoes and long trousers; Avoid touching sprayed crops and report observed chemicals to teacher.

Total Task 2 = 15 marks.

Examiner Guidance for Task 2 (rubric & deductions)

For (a) accept standard biological definitions of biotic and abiotic; if candidates misclassify by confusing living/non-living — deduct 1 mark each.

For (b) allow alternative but correct food chains using observed organisms; if arrows are reversed, deduct 1 mark.

For (d) grade both positive and negative effects; either is acceptable so long as ecological impact is sensible.

Overall Marking Summary (paper)

Task 1: 15 marks

Task 2: 15 marks

Total: 30 marks (but front page shows Task1 15 & Task2 15 — consistent)

Additional Confidential Notes for Examiners (helpful tips)

1. **Consistency checks:** If the candidate's observations contradict the supplied observation in the question (e.g., they claim distilled water produced scum), mark based on the question data and check if they mis-copied; award deductions if they wrote an incorrect conclusion.
2. **Authenticity:** For school-based practicals, encourage the teacher to repeat ambiguous tests (e.g., weak lather) once to confirm.

3. **Time management:** Each practical group should spend ~20–30 minutes on Task 1 (including prep) and ~15 minutes on Task 2 (field observation reporting), rest for cleanup and recording. (Examiner planning note — not to be printed to candidates.)

Quick Reference — What to Record on Mark Sheet (Confidential)

- Candidate name/ASSESSMENT NUMBER , Date, Examiner name.
- For Task 1: tick each of six correct conclusions (6 marks), record two skills (2 marks), three apparatus (3 marks), alternatives for distilled/tap (2 marks), safety (2 marks).
- For Task 2: record classification ticks (5 marks), food chain correctness (2 marks), water support (2 marks), spraying effects (2 marks), animal dependence (2 marks), field safety rules (2 marks)